

SONAL SINGH

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with 2+ years debugging and stabilizing production systems across frontend, backend, and database layers — SIT/UAT, performance testing, live-incident monitoring, and root-cause analysis on a platform architected for 1M+ requests. Proficient in React, Python, and SQL alongside Java/Spring Boot. Direct, hands-on experience building and operating AI/LLM-integrated and agentic systems in production — including a rules-based engine that automatically detects recurring failure patterns — giving a head start on diagnosing and stabilizing AI-generated applications.

WORK EXPERIENCE

Software Engineer

PetroIT Software Solutions

Feb 2024 – Present

Gurugram, India

- Debugged functional and performance issues across API, service, and database layers in a production multitenant SaaS platform (Java/Spring Boot, PostgreSQL) architected for 1M+ requests, tracing root causes from symptom to fix.
- Collaborated with Product and QA teams across SIT, UAT, and performance testing; owned production monitoring and root-cause analysis on live incidents, turning recurring issues into documented fixes for the team.
- Built and maintained production AI-assisted workflows using Claude, OpenAI, and Gemini APIs — including guardrails and validation logic to keep AI-generated outputs stable and correct in a live system.
- Built a secure RBAC framework with Spring Security and optimized 20+ PostgreSQL queries with indexing strategies (40% faster response times), applying the non-happy-path thinking that reliable systems demand.
- Automated build and deployment pipelines using Python scripting and GitHub Actions on AWS (EC2, S3), maintaining a 95% SonarQube code quality score.

PROJECTS

Flight61 — Aviation Safety & Pilot-Training Analytics Platform

2024 – Present

Java 17, Spring Boot 3.3.5, MySQL, Hibernate/JPA, Anthropic Claude API, Zoho CRM, WhatsApp Cloud API

- Built a framework-free rules engine (~19 independent rules, 319 unit tests, ~46,000 LOC) encoding recurring failure patterns from raw telemetry — functioning like an automated troubleshooting playbook that flags and grades issues by severity.
- Orchestrated the analysis pipeline so a single rule's failure can't kill the run, and engineered async-first ingestion (telemetry, logs, nightly CRM sync) on dedicated thread pools so processing never blocks live HTTP requests.
- Integrated the Anthropic Claude API to turn structured findings into plain-English explanations, plus an admin AI chat-bot answering natural-language questions against a read-only DB view — helping non-technical users self-serve answers.

Knowledge-Based Bots — Multi-Agent Engineering Assistant Architecture

2026 – Present

Multi-Agent Orchestration, Vector DB, API Gateway Design, RBAC, System Architecture

- Designing a multi-agent knowledge-bot architecture, each agent scoped to its own domain, tools, and data, unified through a shared orchestration and routing layer to automate reviews, deployments, and documentation.
- Architecting least-privilege, scoped tool access enforced at the middleware layer, and a shared vector memory system for secure cross-agent knowledge sharing, so agents stay correct and can't escalate access.

Anuvaad — AI-Powered Multilingual Enterprise Platform

2024 – Present

Java, Spring Boot, PostgreSQL, Vector DB, OpenAI, Agentic AI, REST APIs

- Built resilient job orchestration across multiple LLM providers (Mistral, LLaMA, Gemma, GPT-class) — automatic retries, rollback, and idempotent re-processing so partial failures never corrupt output, with human-in-the-loop approval before content goes live.
- Built secure external REST APIs and webhook integrations for third-party job submission, with role-based access control and real-time SSE progress tracking.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, Java (Core, OOP, Java 8+)

Frontend & Full Stack: React, Next.js, RESTful APIs, JSON-based integrations

Backend & Frameworks: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Hibernate/ORM, Microservices

Debugging & Reliability: Production support, root-cause analysis, incident RCA, SIT/UAT, performance testing, retries, timeouts, idempotent processing, rollback handling, error handling

Data & Infra: PostgreSQL, MySQL, MongoDB, Redis, Qdrant (Vector DB), Docker, AWS (EC2, S3), GitHub Actions CI/CD

AI / Agentic Systems: LLM integration, RAG pipelines, multi-agent orchestration, prompt engineering, anomaly detection, n8n

Tools: Git, GitHub, Postman, IntelliJ IDEA, VS Code, GitHub Copilot, Cursor, Claude Code

EDUCATION

Bachelor of Engineering, Computer Science

Chandigarh University, Mohali, India

Aug 2020 – Jul 2024